moustaches

Fench statistics (tables and diagrams)

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About

While preparing my lessons, I realized that none of the existing packages for creating statistical diagrams support counts, the French definitions of quartiles, or the construction of histograms and box plots as I teach them… This package was created to fill that gap.

I took inspiration from parts of the LaTeX package ProfCollege (by Christophe Poulain), but the implementation is not a copy, and I did not use AI at all in the development of this package.

Main module: "stat"

- PGCD()
- bandes()
- camembert()
- caracteristiques()
- histogramme()
- simpli()
- stat()

PGCD

Computes the greatest common divisor (GCD) of a list of numbers.

```
#PGCD(1000, 1200, 1400, 1600, 1800,
2000)
```

```
200
```

Parameters

```
PGCD(..a: array) -> int
```

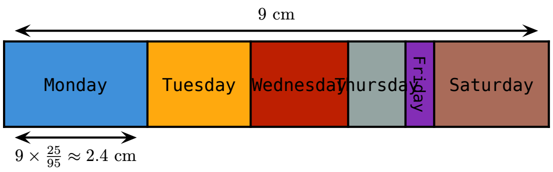
..a `array`

List of numbers

bandes

Draws a band chart.

```
#bandes(
  valeurs:("Monday", "Tuesday",
  "Wednesday", "Thursday", "Friday",
  "Saturday"),
  effectifs:(25, 18, 17, 10, 5, 20),
  width:9cm,
  tourne:(4,),
  explication:(0,),
)
```



Parameters

```
bandes(
  valeurs: array,
  effectifs: array,
  width: length,
  tourne: array,
  couleurs: auto array,
  print: boolean,
  explication: array,
  hauteur: length
) -> content
```

valeurs `array`

Categories (modalities)

Default: `()`

effectifs `array`

Counts of each category

Default: `()`

width `length`

Total width of the chart

Default: `1fr`

tourne `array`

Indices of labels to rotate by 90°

Default: `()`

couleurs `auto` or `array`

List of rectangle colors

Default: `auto`

print `boolean`

If `true`, replaces colors with tiling patterns

Default: `false`

explication `array`

Construction steps to explain, e.g. `(0,)` or `(0,3)`

Default: `()`

hauteur `length`

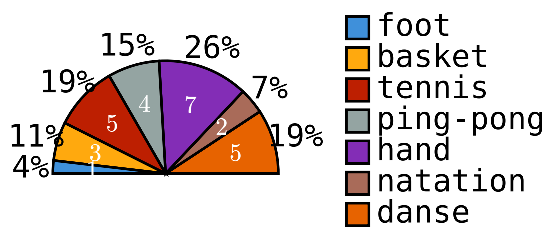
Height of the band chart

Default: `40pt`

camembert

Draws a (semi-)pie chart.

```
#camembert(
  valeurs:
  ("foot","basket","tennis","ping-
  pong","hand","natation","danse"),
  effectifs:(1, 3, 5, 4, 7, 2, 5),
  radius:1.5,
  legende:"d",
  espace:0,
  semi:true,
)
```



Parameters

```
camembert(
  valeurs: array,
  effectifs: array,
  data: array,
  semi: boolean,
  legende: str,
  espace: int float,
  print: boolean,
  couleurs: auto array,
  dedans: boolean none dictionary,
  dehors: boolean none dictionary,
  ..args
) -> content
```

valeurs `array`

Categories (modalities)

Default: `()`

effectifs `array`

Counts of each category

Default: `()`

data `array`

Allows passing a single data array instead

Default: `()`

semi `boolean`

If `true`, draws a semicircular chart

Default: `false`

legende `str`

Legend position: `"d"` (right), `"g"` (left), `"h"` (top), `"b"` (bottom), or `""` for none

Default: `"d"`

espace `int` or `float`

Spacing between legend items when the legend is on the left or right

Default: `.1`

print `boolean`

If `true`, replaces colors with tiling patterns

Default: `false`

couleurs `auto` or `array`

List of sector colors

Default: `auto`

dedans `boolean` or `none` or `dictionary`

`true`, `none`, `false`, or options passed to `inner-label`

Default: `true`

dehors `boolean` or `none` or `dictionary`

`true`, `none`, `false`, or options passed to `outer-label`

Default: `true`

caracteristiques

Computes descriptive statistics and returns them as a dictionary: `effectif-total`, `mediane`, `classe-mediane`, `q1`, `q3`, `d1`, `d9`, `ecart-interquartiles`, `max`, `min`, `etendue`, `modes`, `effectif-modes`, `sommex`, `sommex2`, `moyenne`, `variance`, `ecart-type`, `variance-echantillon`, `ecart-type-echantillon`, `lqboxplot`, and `cetzboxplot`.

```
#caracteristiques(sondage:(1, 5, 7, 9,
4, 5, 9, 8, 7, 3, 8))
```

```
(
  effectif-total: 11,
  mediane: 7,
  classe-médiane: none,
  q1: 4,
  q3: 8,
  d1: 3,
  d9: 9,
  ecart-interquartiles: 4,
  max: 9,
  min: 1,
  etendue: 8,
  modes: (5, 7, 8, 9),
  effectif-modes: 2,
  sommex: 66,
  sommex2: 464,
  moyenne: 6.0,
  variance: 6.181818181818181818,
  ecart-type: 2.4863262420322436,
  variance-echantillon: 6.799999999999997,
  ecart-type-echantillon: 2.607680962081059,
  lqboxplot: (
    median: 7,
    q1: 4,
    q3: 8,
    whisker-low: 3,
    whisker-high: 9,
    outliers: (1,),
    mean: 6.0,
  ),
  cetzboxplot: (
    q1: 4,
    q2: 7,
    q3: 8,
    min: 3,
    max: 9,
    outliers: (1, 6.0),
  ),
)
```

Parameters

```
caracteristiques(
  valeurs: array,
  effectifs: array,
  classes: array,
  bins: int,
  sondage: array,
  crochets: boolean,
  deciles: boolean
) -> dictionary
```

valeurs `array`

Values of the numeric variable

Default: `()`

effectifs `array`

Counts of each value

Default: `()`

classes `array`

Class boundaries, if applicable: length = `effectifs.len() + 1`

Default: `()`

bins `int`

For raw data in `sondage`, automatically creates `bins` classes of equal width

Default: `0`

sondage `array`

List of all observations

Default: `()`

crochets `boolean`

For class intervals, use interval notation (`[a; b[`) instead of inequalities (`a <= ... < b`)

Default: `false`

deciles `boolean`

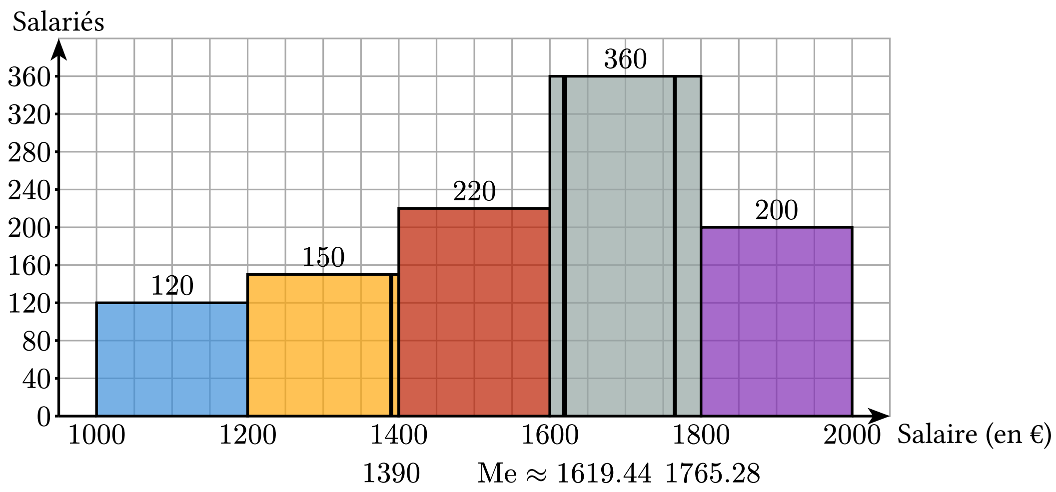
If `true`, whiskers are based on deciles; otherwise they extend to ±1.5 × IQR

Default: `true`

histogramme

Draws a true histogram: class widths do not have to be equal, and rectangle areas are proportional to the counts/frequencies.

```
#histogramme(
   classes: (1000, 1200, 1400, 1600, 1800, 2000),
   effectifs: (120, 150, 220, 360, 200),
   px: 50,
   uaire: 10,
   s: .5,
   // textsize: .5em,
   Donnee: [Salaire (en €)],
   Effectifs: [Salariés],
   Mediane: true,
   quartiles: true,
   pos-rect: (0,5),
 )
```



Parameters

```
histogramme(
  classes: array,
  effectifs: array,
  px: int,
  uaire: int,
  s: int float dictionary,
  grille: boolean,
  PasGrille: int float,
  Donnee: content,
  Effectifs: content,
  add: int array,
  lecture: boolean,
  DonneesSup: boolean,
  ListeCouleurs: auto color array,
  pos-rect: array,
  print: boolean,
  transparence: ratio,
  ECC: boolean,
  Mediane: boolean,
  quartiles: boolean,
  décimales: int,
  posq1: array,
  posMediane: array,
  posq3: array,
  stroke-mediane: length stroke,
  stroke-quartiles: length stroke,
  textsize: length
) -> content
```

classes `array`

Class boundaries: length = `effectifs.len() + 1`

Default: `()`

effectifs `array`

Counts of each class: length = `classes.len() - 1`

Default: `()`

px `int`

Tick interval on the x-axis

Default: `1`

uaire `int`

Quantity represented by one grid square

Default: `1`

s `int` or `float` or `dictionary`

Scale passed to cetz

Default: `1`

grille `boolean`

Whether to display a grid

Default: `true`

PasGrille `int` or `float`

Grid spacing

Default: `1`

Donnee `content`

Label for the legend or y-axis, if applicable

Default: `[Values]`

Effectifs `content`

Label for the x-axis

Default: `[Frequencies]`

add `int` or `array`

Extra margin on the left, right, and bottom. Either a single integer or a triple

Default: `(-1, 1, 1)`

lecture `boolean`

Whether to display a y-axis when all classes have the same width

Default: `true`

DonneesSup `boolean`

Whether to display frequencies above the rectangles

Default: `true`

ListeCouleurs `auto` or `color` or `array`

Rectangle colors, or a single color (e.g. `white` for printing)

Default: `auto`

pos-rect `array`

Position of the legend rectangle relative to the upper-left corner of the smallest rectangle

Default: `(0, 2)`

print `boolean`

If `true`, replaces colors with tiling patterns

Default: `false`

transparence `ratio`

Rectangle transparency. Use `100%` if `ListeCouleurs: white`

Default: `30%`

ECC `boolean`

Displays cumulative frequencies

Default: `false`

Mediane `boolean`

Displays the median

Default: `false`

quartiles `boolean`

Displays the quartiles

Default: `false`

décimales `int`

Number of decimal places for rounded values

Default: `2`

posq1 `array`

Position of the Q1 label relative to the x-axis

Default: `(0,-1.5)`

posMediane `array`

Position of the median label relative to the x-axis

Default: `(0,-1.5)`

posq3 `array`

Position of the Q3 label relative to the x-axis

Default: `(1,-1.5)`

stroke-mediane `length` or `stroke`

Stroke used to draw the median

Default: `2pt`

stroke-quartiles `length` or `stroke`

Stroke used to draw the quartiles

Default: `1.5pt`

textsize `length`

Text size

Default: `1em`

simpli

Writes the quotient `a / b`, where `a` and `b` are floating-point numbers (or simply `a` if `b = 1`), together with its reduced fractional form (up to 6 decimal places).

```
#simpli(3.6,6.9)
```

$$\frac{3.6}{6.9} = \frac{12}{23}$$

Parameters

```
simpli(
  a,
  b
) -> content
```

a

int | float

b

int | float

stat

Performs a statistical analysis and displays the results as a table, with optional frequencies (`effectifs`), relative frequencies (`"f"` for fractions, `"d"` for decimals, `"t"` for all, `true` for percentages), sector angles for (semi-)pie charts, cumulative frequencies (`ECC`, `FCC`), class intervals (automatically generated when `bins > 0`) and their centers.

It can also display charts: `"hbar"`, `"bar"`, `"circ"`, `"semicirc"`, `"box"`, `"bande"`, `"histo"`, or any combination of these.

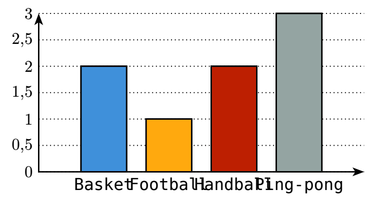
If `multi != ()`, grouped bar charts (horizontal or vertical) or multiple box plots can be displayed. Setting `print: true` replaces colors with tiling patterns.

```
#stat(
  classes:(150, 160, 170, 200),
  effectifs:(3, 4, 5),
  frequences:"f",
)
```

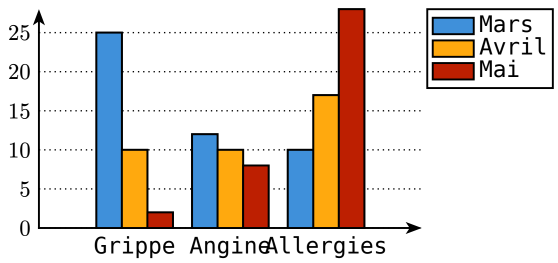
	$150 \leqslant ... < 160$	$160 \leqslant ... < 170$	$170 \leqslant ... < 200$
Centres des classes	155	165	185
Effectifs	3	4	5
Fréquences	$\frac{3}{12} = \frac{1}{4}$	$\frac{4}{12} = \frac{1}{3}$	$\frac{5}{12}$

```
#stat(
  sondage:
  ("Handball","Basket","Football","Handba
          "Ping-pong","Basket","Ping-
          pong","Ping-pong"),
  diagramme:"bar",
  cbar:(size:(8.2,4))
)
```

	Basket	Football	Handball	Ping-pong
Effectifs	2	1	2	3
Fréquences (en %)	25	13	25	38



```
#stat(
  valeurs:
  ("Grippe","Angine","Allergies"),
  multi:((25,12,10),(10,10,17),
  (2,8,28)),
  diagramme:"bar",
  labels:("Mars","Avril","Mai"),
  tableau:false
)
```



Parameters

```
stat(
  valeurs: array,
  effectifs: array,
  qualitatif: boolean,
  totaux: boolean,
  tableau: boolean,
  couleur-tableau: color,
  inset: length,
  Nom-donnee: content,
  Nom-effectifs: content,
  décimales: int,
  frequences: boolean str,
  sondage: array,
  vide: boolean,
  angle: boolean str,
  ECC: boolean str,
  classes: array,
  bins: int,
  crochets: boolean,
  centre: boolean str,
  colonnes-vide: array,
  cases-vide: array,
  diagramme: str,
  bar: array,
  cbar: array,
  moustaches: array,
  cmoustaches: array,
  circ: dictionary,
  bande: array,
  histo: array,
  FCC: boolean str,
  multi: array,
  labels: array,
  print: boolean,
  ListeCouleurs: auto color array
) -> content
```

valeurs `array`

Values or categories of the variable

Default: `()`

effectifs `array`

Counts of each value

Default: `()`

qualitatif `boolean`

Whether the variable is qualitative

Default: `true`

totaux `boolean`

Display totals in tables

Default: `false`

tableau `boolean`

Display the table

Default: `true`

couleur-tableau `color`

Background color of the first row and first column

Default: `luma(85%)`

inset `length`

Cell inset

Default: `5pt`

Nom-donnee `content`

Label for the upper-left table cell (and some charts)

Default: `[]`

Nom-effectifs `content`

Label for the frequency row (and some charts)

Default: `[Frequencies]`

décimales `int`

Number of decimal places for rounded values

Default: `0`

frequences `boolean` or `str`

Relative frequencies: `true` for percentages, `"f"` for fractions, `"d"` for decimals, `"t"` for all, `"v"` for an empty row

Default: `true`

sondage `array`

List of all observations

Default: `()`

vide `boolean`

If `true`, all rows except the first are left blank

Default: `false`

angle `boolean` or `str`

Display sector angles (`"v"` for an empty row)

Default: `false`

ECC `boolean` or `str`

Display cumulative frequencies (`"v"` for an empty row)

Default: `false`

classes `array`

Class boundaries for continuous data: length = `effectifs.len() + 1`

Default: `()`

bins `int`

For raw data, automatically creates `bins` classes of equal width

Default: `0`

crochets `boolean`

Use interval notation (`[a; b[`) instead of inequalities (`a <= ... < b`)

Default: `false`

centre `boolean` or `str`

Display class centers (`"v"` for an empty row)

Default: `true`

colonnes-vide `array`

Columns to leave empty

Default: `()`

cases-vide `array`

Individual cells to leave empty, numbered row by row

Default: `()`

diagramme `str`

Chart type: `"hbar"`, `"bar"`, `"circ"`, `"semicirc"`, `"box"`, `"bande"`, `"histo"`, or any combination

Default: `""`

bar `array`

Parameters passed to the horizontal/vertical bar chart canvas

Default: `()`

cbar `array`

Parameters passed to cetz's `barchart` or `columnchart`

Default: `()`

moustaches `array`

Parameters passed to the custom box-plot function

Default: `()`

cmoustaches `array`

Parameters passed to cetz's `boxwhiskers`

Default: `()`

circ `dictionary`

Parameters passed to `camembert`

Default: `(:)`

bande `array`

Parameters passed to `bandes`

Default: `()`

histo `array`

Parameters passed to `histogramme`

Default: `()`

FCC `boolean` or `str`

Display cumulative relative frequencies (`"v"` for an empty row)

Default: `false`

multi `array`

Multiple series for grouped bar charts or multiple box plots

Default: `()`

labels `array`

Labels for the different series

Default: `()`

print `boolean`

If `true`, replaces colors with tiling patterns

Default: `false`

ListeCouleurs `auto` or `color` or `array`

Colors used in the charts

Default: `auto`

Sub-module for tilings and colormaps: “hachures”

- alveoles()
- cross()
- dots()
- etoile()
- grille()
- nelines()
- nwlines()
- stripes()

Variables

- hachures
- typst18
- cycles-couleurs

alveoles

hexagons tiling

Parameters

```
alveoles(
  c: color,
  size: length,
  background: color,
  fill: none color
) -> tiling
```

cross

colored cross tiling that print in b&w

Parameters

```
cross(
  c: color,
  size: length,
  fill: auto color
) -> tiling
```

dots

colored dots tiling that print in b&w

Parameters

```
dots(
  c: color,
  size: length,
  fill: auto color
) -> tiling
```

etoile

stars tiling

Parameters

```
etoile(
  c: color,
  size: length,
  background: color,
  fill: none color
) -> tiling
```

grille

grid lines tiling

Parameters

```
grille(
  c: color,
  size: length,
  fill: color
) -> tiling
```

nelines

North-east lines tiling

Parameters

```
nelines(
  c: color,
  size: length,
  fill: color
) -> tiling
```

nwlines

North-west lines tiling

Parameters

```
nwlines(
  c: color,
  size: length,
  fill: color
) -> tiling
```

stripes

colored stripes tiling that print in b&w

Parameters

```
stripes(
  c: color,
  size: length,
  fill: auto color
) -> tiling
```

hachures `array`

List of tilings

typst18 `array`

colormap of basic typst colors

cycles-couleurs `array`

25 cycles de couleurs disponibles